

Lanka Tool Hire - Team Git Branching & Commit Workflow Master Guide

Module: COMP70066 Software Engineering Principles and Practices

Lab Reference: Lab 01 - Setting Up Git and GitHub (Three-Lens Framework: Mechanism, Responsibility, Policy)

Project: Lanka Tool Hire (Pvt) Ltd (Shelton Tool Hire) Review & Hire Portal

Repository: <https://github.com/hashan-lakshitha/lanka-tool-hire-portal.git>

Target Duration: 8 Sprints (10 Aug 2026 – 04 Oct 2026)

1. Team Members & Roles

Member Name	Student ID	GitHub Responsibility & Focus Area
Hashan Lakshitha Walauwatta	CB018777	Repo Owner & Initial Scaffold, Password Reset, Reviews/Ratings, Moderation, Localization (i18n)
Abdullah Suhail	CB010327	Scrum Master, Auth/Security, Navigation & Core Layouts, Rental Cost Calculator, Releases & Tags
Mohamed Shadil	CB018738	Database Migrations & Sequelize Models, Image/Media Upload API, Inventory Checks, Admin Analytics

2. Theoretical Framework (Lab 01 Three-Lens Alignment)

Lens	Question Answered	Implementation in This Project
Mechanism	How changes are physically captured and stored	<code>git commit</code> , <code>git push</code> , branch pointers, Git object store, tags (<code>v0.1</code> - <code>v0.8</code>)
Responsibility	Who is accountable for each step	Distinct commit authors (<code>user.name</code> , <code>user.email</code>), PR reviews and approvals
Policy	Rules governing the mechanism	Conventional Commits (<code>feat:</code> , <code>fix:</code> , <code>chore:</code> , <code>docs:</code>), branch naming (<code>feature/...</code>), merge to <code>dev</code> only

3. Initial Git Identity Setup (Run Once per Machine)

Configure your local Git identity before creating or pushing commits:

Hashan Walauwatta:

```
bash
git config user.name "Hashan Lakshitha Walauwatta"
git config user.email "hashan@example.com"
```

Abdullah Suhail:

```
bash
git config user.name "Abdullah Suhail"
git config user.email "suhail@example.com"
```

Mohamed Shadil:

```
bash
git config user.name "Mohamed Shadil"
git config user.email "shadil@example.com"
```

4. Phase 0: Project Initial Scaffolding (Hashan Walauwatta)

Repository link: <https://github.com/hashan-lakshitha/lanka-tool-hire-portal.git>

```
bash
# Ensure you are in lanka-tool-hire-portal repository
git checkout main
git pull origin main

# Create and switch to integration branch
git checkout -b dev

# Copy project configuration and core dependencies from shelton-tools:
# .gitignore, package.json, package-lock.json, next.config.mjs, postcss.config.mjs,
jsconfig.json, eslint.config.mjs, .env.example, .sequelizerc
# Base layouts: src/app/globals.css, src/app/layout.js, src/app/page.js,
src/app/favicon.ico, src/app/apple-icon.png
```

```
git add .gitignore package.json package-lock.json next.config.mjs postcss.config.mjs
jsconfig.json eslint.config.mjs .env.example .sequelizerc
git add src/app/globals.css src/app/layout.js src/app/page.js src/app/favicon.ico
src/app/apple-icon.png

git commit -m "chore: initialize Next.js 14 project scaffold with Tailwind CSS and
base configs"
git push -u origin dev
```

5. Phase 0.1: Repository Synchronization (Suhail & Shadil)

Once Hashan has pushed `dev`, Suhail and Shadil should pull/switch to `dev`:

```
bash
git clone https://github.com/hashan-lakshitha/lanka-tool-hire-portal.git
cd lanka-tool-hire-portal
git checkout dev
git pull origin dev
```

6. Sprint-by-Sprint Execution Workflow (Sprints 1 to 8)

Sprint 1 (Aug 10 - Aug 16): Account & Staff Foundations (Epics A, M, O)

1. Mohamed Shadil:

```
bash
git checkout dev && git pull origin dev
git checkout -b feature/A-database-setup

# Stage DB configuration, migrations, and User model
git add src/config/ src/lib/db.js migrations/20240101000001-create-users.js
src/models/user.js src/models/index.js
git commit -m "feat(db): setup Sequelize connection and User model with migration"
git push -u origin feature/A-database-setup
git checkout dev && git merge feature/A-database-setup && git push origin dev
```

2. Abdullah Suhail:

```
bash
```

```
git checkout dev && git pull origin dev
git checkout -b feature/A-auth-system

# Stage NextAuth, registration API, navigation, and auth pages
git add src/lib/auth.js src/middleware.js src/app/api/auth/[...nextauth]/
src/app/api/auth/register/ src/app/providers.js
git add src/components/Nav.js src/app/login/ src/app/register/
git commit -m "feat(auth): integrate NextAuth authentication and user registration
portal"
git push -u origin feature/A-auth-system
git checkout dev && git merge feature/A-auth-system && git push origin dev
```

3. Hashan Walauwatta:

```
bash

git checkout dev && git pull origin dev
git checkout -b feature/A-password-reset

# Stage mailer utility and password reset pages/routes
git add src/lib/mailer.js src/app/forgot-password/ src/app/reset-password/
src/app/api/auth/forgot-password/ src/app/api/auth/reset-password/
src/app/api/auth/change-password/
git commit -m "feat(auth): implement password reset workflow with email notification
support"
git push -u origin feature/A-password-reset
git checkout dev && git merge feature/A-password-reset && git push origin dev
```

Sprint 1 Release (Abdullah Suhail):

```
bash

git checkout main && git pull origin main
git merge dev
git tag -a v0.1 -m "Release v0.1: User authentication, database foundations, and
base layout"
git push origin main --tags
```

Sprint 2 (Aug 17 - Aug 23): Equipment Catalogue & Media (Epics J, L)

1. Mohamed Shadil:

```
bash

git checkout dev && git pull origin dev
git checkout -b feature/J-catalogue-models
```

```
git add migrations/20240101000003-create-categories.js migrations/20240101000004-
create-tools.js src/models/category.js src/models/tool.js src/app/api/admin/upload/
git commit -m "feat(models): add Category and Tool schemas with image upload API"
git push -u origin feature/J-catalogue-models
git checkout dev && git merge feature/J-catalogue-models && git push origin dev
```

2. Abdullah Suhail:

```
bash
git checkout dev && git pull origin dev
git checkout -b feature/J-admin-categories

git add src/app/admin/layout.js src/app/admin/page.js src/app/admin/login/
src/app/admin/categories/ src/app/api/admin/categories/ src/models/adminUser.js
migrations/20240101000002-create-admin-users.js src/lib/adminStyles.js
git commit -m "feat(admin): build admin portal layout and category management
module"
git push -u origin feature/J-admin-categories
git checkout dev && git merge feature/J-admin-categories && git push origin dev
```

3. Hashan Walauwatta:

```
bash
git checkout dev && git pull origin dev
git checkout -b feature/L-equipment-management

git add src/app/admin/equipment/ src/app/api/admin/tools/
git commit -m "feat(admin): create tool inventory and equipment management UI"
git push -u origin feature/L-equipment-management
git checkout dev && git merge feature/L-equipment-management && git push origin dev
```

Sprint 2 Release (Abdullah Suhail):

```
bash
git checkout main && git pull origin main
git merge dev
git tag -a v0.2 -m "Release v0.2: Equipment catalogue, admin management, and media
upload"
git push origin main --tags
```

Sprint 3 (Aug 24 - Aug 30): Pricing Rules & Browsing (Epics K, B)

1. Mohamed Shadil:

```
bash
```

```
git checkout dev && git pull origin dev
git checkout -b feature/K-public-catalogue-api

git add src/app/api/tools/route.js src/app/api/categories/route.js
git commit -m "feat(api): create public catalogue listing endpoints with filtering support"
git push -u origin feature/K-public-catalogue-api
git checkout dev && git merge feature/K-public-catalogue-api && git push origin dev
```

2. Abdullah Suhail:

```
bash
```

```
git checkout dev && git pull origin dev
git checkout -b feature/B-catalogue-ui

git add src/components/ToolCard.js src/app/catalogue/page.js src/lib/theme.js
git commit -m "feat(ui): design responsive ToolCard and catalogue browsing page"
git push -u origin feature/B-catalogue-ui
git checkout dev && git merge feature/B-catalogue-ui && git push origin dev
```

3. Hashan Walauwatta:

```
bash
```

```
git checkout dev && git pull origin dev
git checkout -b feature/B-category-filters

git add src/app/catalogue/page.js
git commit -m "feat(ui): add category sidebar filter and price sorting controls"
git push -u origin feature/B-category-filters
git checkout dev && git merge feature/B-category-filters && git push origin dev
```

Sprint 3 Release (Abdullah Suhail):

```
bash
```

```
git checkout main && git pull origin main
git merge dev
git tag -a v0.3 -m "Release v0.3: Public catalogue browsing, card components, and filtering"
git push origin main --tags
```

Sprint 4 (Aug 31 - Sep 06): Detail Pages & Search Engine (Epics D, C)

1. Mohamed Shadil:

```
bash
```

```
git checkout dev && git pull origin dev
git checkout -b feature/C-inventory-checker

git add src/lib/inventory.js src/app/api/tools/[id]/route.js
git commit -m "feat(inventory): add availability check helper and single tool lookup API"
git push -u origin feature/C-inventory-checker
git checkout dev && git merge feature/C-inventory-checker && git push origin dev
```

2. Abdullah Suhail:

```
bash
```

```
git checkout dev && git pull origin dev
git checkout -b feature/D-tool-detail-page

git add src/app/tools/[id]/page.js
git commit -m "feat(ui): create dynamic tool specifications and availability detail view"
git push -u origin feature/D-tool-detail-page
git checkout dev && git merge feature/D-tool-detail-page && git push origin dev
```

3. Hashan Walauwatta:

```
bash
```

```
git checkout dev && git pull origin dev
git checkout -b feature/C-search-engine

git add src/components/Nav.js src/app/catalogue/page.js
git commit -m "feat(search): add instant debounced search across equipment catalogue"
git push -u origin feature/C-search-engine
git checkout dev && git merge feature/C-search-engine && git push origin dev
```

Sprint 4 Release (Abdullah Suhail):

```
bash
```

```
git checkout main && git pull origin main
git merge dev
git tag -a v0.4 -m "Release v0.4: Tool details page, search bar, and inventory verification"
git push origin main --tags
```

Sprint 5 (Sep 07 - Sep 13): Rental Cost Calculator (Epic E)

1. Mohamed Shadil:

```
bash

git checkout dev && git pull origin dev
git checkout -b feature/E-rental-models

git add migrations/20240101000007-create-rental-quotes.js migrations/20240101000009-
create-rentals-and-tool-quantity.js src/models/rentalQuote.js src/models/rental.js
src/app/api/rentals/ src/app/api/customer/rentals/
git commit -m "feat(models): implement RentalQuote and Rental entities with database
migrations"
git push -u origin feature/E-rental-models
git checkout dev && git merge feature/E-rental-models && git push origin dev
```

2. Abdullah Suhail:

```
bash

git checkout dev && git pull origin dev
git checkout -b feature/E-hire-calculator

git add src/components/HireCalculator.js src/app/api/quote/route.js
git commit -m "feat(calc): build interactive HireCalculator with deposit and
duration logic"
git push -u origin feature/E-hire-calculator
git checkout dev && git merge feature/E-hire-calculator && git push origin dev
```

3. Hashan Walauwatta:

```
bash

git checkout dev && git pull origin dev
git checkout -b feature/E-customer-quotes

git add src/app/dashboard/page.js src/app/dashboard/quotes/
src/app/dashboard/rentals/ src/app/api/customer/quotes/ src/app/profile/
src/app/api/customer/profile/
git commit -m "feat/dashboard): add customer dashboard for managing quotes and
active rentals"
git push -u origin feature/E-customer-quotes
git checkout dev && git merge feature/E-customer-quotes && git push origin dev
```

Sprint 5 Release (Abdullah Suhail):

```
bash
```



```
git checkout main && git pull origin main
git merge dev
git tag -a v0.5 -m "Release v0.5: Rental cost calculator, quote generation, and
customer portal"
git push origin main --tags
```

Sprint 6 (Sep 14 - Sep 20): Reviews, Ratings & Comments (Epics F, G)

1. Mohamed Shadil:

```
bash
git checkout dev && git pull origin dev
git checkout -b feature/F-review-models

git add migrations/20240101000005-create-reviews.js migrations/20240101000008-
create-review-comments.js src/models/review.js src/models/reviewComment.js
git commit -m "feat(models): add Review and ReviewComment schemas with relational
associations"
git push -u origin feature/F-review-models
git checkout dev && git merge feature/F-review-models && git push origin dev
```

2. Hashan Walauwatta:

```
bash
git checkout dev && git pull origin dev
git checkout -b feature/F-rating-components

git add src/lib/rating.js src/components/StarRating.js src/components/ReviewList.js
src/app/api/reviews/[id]/comments/
git commit -m "feat(ui): implement StarRating and ReviewList with Bayesian score
calculations"
git push -u origin feature/F-rating-components
git checkout dev && git merge feature/F-rating-components && git push origin dev
```

3. Abdullah Suhail:

```
bash
git checkout dev && git pull origin dev
git checkout -b feature/F-review-submission

git add src/components/ReviewForm.js src/app/api/reviews/route.js
src/app/dashboard/reviews/ src/app/api/customer/reviews/
git commit -m "feat(reviews): build ReviewForm submission and customer review
```

```
history"
git push -u origin feature/F-review-submission
git checkout dev && git merge feature/F-review-submission && git push origin dev
```

Sprint 6 Release (Abdullah Suhail):

```
bash
git checkout main && git pull origin main
git merge dev
git tag -a v0.6 -m "Release v0.6: Customer reviews, star ratings, and community comments"
git push origin main --tags
```

Sprint 7 (Sep 21 - Sep 27): Company Responses & Moderation (Epics H, I, J5)

1. Mohamed Shadil:

```
bash
git checkout dev && git pull origin dev
git checkout -b feature/H-company-response-model

git add migrations/20240101000006-create-company-responses.js
src/models/companyResponse.js
git commit -m "feat(models): create CompanyResponse entity and database migrations"
git push -u origin feature/H-company-response-model
git checkout dev && git merge feature/H-company-response-model && git push origin dev
```

2. Hashan Walauwatta:

```
bash
git checkout dev && git pull origin dev
git checkout -b feature/I-admin-moderation

git add src/app/admin/moderation/ src/app/api/admin/reviews/route.js
git commit -m "feat(admin): build moderation console for approving or rejecting customer reviews"
git push -u origin feature/I-admin-moderation
git checkout dev && git merge feature/I-admin-moderation && git push origin dev
```

3. Abdullah Suhail:

```
bash
```

```
git checkout dev && git pull origin dev
git checkout -b feature/H-official-response

git add src/app/api/admin/reviews/[id]/response/route.js src/app/admin/rentals/
src/app/api/admin/rentals/
git commit -m "feat(admin): implement official staff response API and admin rentals
management"
git push -u origin feature/H-official-response
git checkout dev && git merge feature/H-official-response && git push origin dev
```

Sprint 7 Release (Abdullah Suhail):

```
bash

git checkout main && git pull origin main
git merge dev
git tag -a v0.7 -m "Release v0.7: Review moderation workflow and official company
responses"
git push origin main --tags
```

Sprint 8 (Sep 28 - Oct 04): Reporting, Testing & Release (Epics N, P)

1. Hashan Walauwatta:

```
bash

git checkout dev && git pull origin dev
git checkout -b feature/P-i18n-localization

git add src/lib/i18n.js src/components/LanguageSwitcher.js src/locales/
git commit -m "feat(i18n): implement trilingual localization support (EN, SI, TA)"
git push -u origin feature/P-i18n-localization
git checkout dev && git merge feature/P-i18n-localization && git push origin dev
```

2. Mohamed Shadil:

```
bash

git checkout dev && git pull origin dev
git checkout -b feature/N-admin-analytics

git add src/app/admin/analytics/ src/app/api/admin/analytics/ public/
git commit -m "feat(analytics): implement admin analytics dashboard with revenue
charts"
git push -u origin feature/N-admin-analytics
git checkout dev && git merge feature/N-admin-analytics && git push origin dev
```

3. Abdullah Suhail:

```
bash
git checkout dev && git pull origin dev
git checkout -b feature/release-polish

git add script/ README.md
git commit -m "docs: finalize deployment instructions and seed scripts for
production"
git push -u origin feature/release-polish
git checkout dev && git merge feature/release-polish && git push origin dev
```

Sprint 8 Final Release (Abdullah Suhail):

```
bash
git checkout main && git pull origin main
git merge dev
git tag -a v0.8 -m "Release v0.8: Final release candidate with analytics and full
localization"
git push origin main --tags
```

7. Best Practices & Academic Guidelines

- **Zero Secrets Policy:** Never commit `.env` files containing actual passwords or DB credentials. Only commit `.env.example`.
- **Pre-Branch Synchronization:** Always execute `git checkout dev && git pull origin dev` before initiating a new feature branch.
- **Conventional Commits:** Ensure commit messages strictly adhere to standard formats (`feat:`, `fix:`, `chore:`, `docs:`).
- **Fast-Forward & Merge Integrity:** Ensure feature branches merge cleanly into `dev`.